PROJECT 2: CONCRETE COMPRESSIVE STRENGTH

Predicting strength from mixture composition and curing age

1. Executive summary

This project predicts concrete compressive strength in megapascals (MPa) using eight mixture and curing-age variables. The dataset contains 1,030 observations and has no missing values. Four regression approaches were compared with a training-mean baseline. Histogram gradient boosting achieved the best held-out test performance: MAE 3.08 MPa, RMSE 4.35 MPa and R^2 0.9264. The results show that nonlinear models outperform the linear model on this dataset, but they should be treated as predictive associations rather than causal evidence.

2. Data understanding and preparation

The predictors are cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate and age (days). The target is concrete compressive strength (MPa). Each of the nine columns has 1,030 non-missing observations. The target mean is 35.82 MPa, standard deviation 16.71 MPa, minimum 2.33 MPa and maximum 82.60 MPa.

All predictors are quantitative and have direct engineering meaning, so all were retained. No categorical encoding was required. A median imputer was included in each model pipeline for reproducibility and future robustness, although no missing cells were present. Standardisation was applied to the linear regression pipeline; tree-based models were used without scaling. An 80% training / 20% test split with random_state=42 was used.

3. Baseline and model comparison

The mean baseline predicts the training-set mean for every test observation. It produced MAE 13.05 MPa, RMSE 16.05 MPa and R^2 approximately 0, providing a reference point. Lower MAE/RMSE and higher R^2 indicate better predictive performance.

Model	MAE (MPa)	RMSE (MPa)	R^2
Mean baseline	13.052	16.054	-0.0002
Linear regression	7.745	9.797	0.6275
Random forest	4.032	5.686	0.8745
Extra Trees	3.659	5.355	0.8887
HistGradientBoosting	3.079	4.354	0.9264

The best-performing model was HistGradientBoosting. Its RMSE was about 72.9% lower than the mean baseline RMSE and its R^2 indicates that it explained approximately 92.6% of the variation in the held-out test targets. Extra Trees and Random Forest also substantially outperformed linear regression, supporting the presence of nonlinearities and/or interactions between mixture variables.

4. Residual analysis and weakest predictions

Residuals were defined as actual minus predicted strength. The notebook plots residuals against fitted values, a residual histogram and actual-versus-predicted values. It also lists the ten test observations with the largest absolute prediction errors. For the selected model, the largest absolute error was approximately 15.72 MPa; the next two were 15.71 MPa and 14.81 MPa. These errors are materially larger than the model's average absolute error of 3.08 MPa and show that performance is not uniform across all mixtures.

The weakest predictions should be inspected for unusual combinations of cement, supplementary cementitious materials, water, superplasticizer, aggregates and curing age. Residual patterns may also indicate heteroscedasticity or missing variables. The worst-error observations are evidence that even a high overall R^2 does not guarantee accuracy for every batch.

5. Does cement-strength association prove a real-world effect?

No. A predictive association between cement content and strength does not establish the causal effect of increasing cement. The dataset is observational, and cement content changes alongside water, slag, fly ash, superplasticizer, aggregate proportions and curing age. These variables may confound or modify the observed association. The model learns patterns among mixtures represented in the dataset; it does not automatically estimate what would happen under an intervention that changes cement while holding other conditions constant.

A causal answer requires a controlled experimental programme or a defensible causal design. For example, prepare replicated batches in which cement content is deliberately varied while other ingredients, mixing procedure, curing conditions and testing procedures are controlled or systematically randomised. Predictions outside the observed data range are extrapolations and need separate experimental validation.

6. Conclusion and responsible use

Histogram gradient boosting is the strongest model among those compared on the fixed test split. It should be considered a decision-support or screening tool, not a substitute for standard laboratory compressive-strength tests, engineering judgement or construction regulations. The dataset may not represent every cement source, aggregate type, climate, curing regime or production process. Independent validation and uncertainty assessment are required before practical engineering deployment.

Reproducibility: the accompanying Jupyter notebook includes exploration, baseline, model comparison, metrics, residual plots, worst-error inspection and causal interpretation. Install dependencies in requirements.txt and run all cells from the project folder.